Arjun Madaan

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Coder. Communicator. Lifelong Learner.

RECENT EXPERIENCES

Stanford University, *Stanford Math Circle*

JANUARY 2026 — MARCH 2026 (STANFORD, CA)

Weekly sessions in High School (a.k.a. Advanced) Math Circle. Engaged in sessions led by experts in the field of mathematics. Studied skills and topics in higher mathematics such as Finite State Automata and Graph Theory.

Brown University, *Smartphone and Computational Physics*

JULY 2025, PROVIDENCE, RI

Conducted various physics experiments at Brown University, studying solid mechanics, wave physics, electromagnetism, optics, kinematics, motion (1D & 2D, rotational, periodic, circular), forces, momentum, energy, thermodynamics, conservation laws, and modern physics concepts. Used computational techniques on projectile motion. Worked with unsupervised ML and data visualization softwares. Presented on optimizing Generative AI hyperparameters for accuracy on complex physics problems.

Stanford University - Center for Artificial Intelligence in Medicine & Imaging, *Summer Health AI Bootcamp*

JUNE 2025, ONLINE (STANFORD, CA)

Studied applications of AI in healthcare. Covered essential concepts and principles, opportunities and challenges of health AI, foundational language models for healthcare, ethical AI practices, and dataset curation for health AI. Engaged in lectures, tutorials, journal clubs, and expert sessions, and coded ML models for healthcare.

PyData Virginia, *Session Chair*

APRIL 2025, CHARLOTTESVILLE, VA

Chaired sessions at the PyData Virginia conference. Topics included GPU Accelerated Data Science given by NVIDIA®, Artificial Intelligence in SciPy, and the use of Python to study HIV health disparities.

RECENT PROJECTS

M.C.A.I.T. (Medical and Cross-disciplinary AI Test)

FEBRUARY 2025

Comprehensive, ten-task Kaggle benchmark suite evaluating LLM capability and safety for medicine, assessing ethics, scientific knowledge, patient interaction, and diagnostic ability. Made custom tasks using numerical scoring over binary scores for comprehensive evaluations.

Gemma 3n for Clinical Diagnosis Support, *Submission to Google DeepMind Gemma 3n Impact Challenge*

JULY-AUGUST 2025

On-device program using Gemma 3n for diagnosis. Had information-dense JSONs (few-shot prompts), mandatory ethical disclaimers, and Gradio GUI.

TEST SCORES

PSAT 8/9® - 1410 (**99th percentile**), Math - 720/720, EBRW - 690/720

PERFORMANCE REPORTS

Professor Renuka Rajapakse (University of Massachusetts Dartmouth) - "Arjun and his group **presented an excellent evaluation** of how successful Generative AI was in solving complex physics problems. Arjun **took the lead** in selecting the topic and guiding his fellow group members. The information presented was **very useful for me** as a teacher of physics, and I am **grateful to Arjun** and his group for taking on this subject for their research project." **Quoted from Smartphone and Computational Physics Performance Report**

SKILLS

Machine Learning & Generative AI (including RAG)

Prompt Engineering

Explainable AI ("X-AI")

RECOGNITION

Silver Medal, British Biology Olympiad, United Kingdom Biology Competitions

Best Delegate, Governor's School Model United Nations Conference XXVII (Special Political & Decolonization Committee)

Bronze Medal, Kaggle Notebook (Mental Health Diagnosis)

Bronze Medal, Kaggle Notebook (FinAI)